ThermoBench: Technical Submission Report

Candidate ID: RA2311026010405 | **Date:** October 31, 2026 | **Project:** Windows 11 x64 Engineering Workbench

🔗 Official Submission & Evaluation Links

🎬 **Video Walkthrough (3-5 min):**
Watch Video on Google Drive

🌐 **Live Web Deployment:**
https://3d-varsha.vercel.app/

💻 **GitHub Source Repository:**
https://github.com/var-13/3D

📦 **Version Control & Artifacts:**
61/61 Unit Tests Passing · Git Bundle Included

1. Executive Summary & Architecture Choices

ThermoBench is a high-performance engineering workbench for thermal FEA, multicore parallel evaluation, neural network backpropagation, 3-D geometry parsing, and embedded hardware simulation. It enforces strict separation of concerns:

- **Pure C++17 Core (`src/core/`)**: Decoupled numerical libraries with zero graphics/network dependencies. Implements Thomas tridiagonal algorithm (TDMA, *O(N)* complexity), backpropagation with analytical gradients, and atomic multicore dispatch.
- **Micro-Server Layer (`src/server/`)**: Lightweight HTTP daemon using `cpp-httplib` with zero external link dependencies, exposing RESTful endpoints for geometry, solver executions, and hardware telemetry.
- **Offline Web Frontend (`ui/`)**: Responsive single-page interface with Three.js r167 3D WebGL CAD/FEA viewer and Chart.js 4.4.3. All vendor assets are bundled locally for 100% offline and air-gapped capability.
- **Process Supervisor (`Launch.exe`)**: Native Win32 launcher that verifies server health over non-blocking TCP sockets, auto-launches the system browser, and cleanly manages child process lifecycles.

2. 1-D Thermal FEA Verification & Analytical Superconvergence

The steady-state 1-D heat conduction equation with area A, thermal conductivity k, and internal volumetric generation q is:
$-(d/dx)(k\,A\,dT/dx) = q\,A$, with boundary conditions $T(0) = T_0 = 100\ °C$, $T(L) = T_L = 20\ °C$ on $L = 1.0\ m$, $A = 10^{-4}\ m^2$, $k = 10\ W/(m·K)$, $q = 10^4\ W/m^3$.

Analytical Exact Solution: $T_{exact}(x) = T_0 + ((T_L - T_0)/L)x + (q / (2k))\,x\,(L - x)$. The global tridiagonal system is assembled using 2-node linear elements and solved via the Thomas algorithm:

Mesh Density N	Element Size h (m)	Midpoint Error at x=0.5 m (°C)	Theoretical $125 \cdot h^2$ (°C)	Observed Ratio	Nodal Superconvergence Error
N = 10	0.1000	1.250000	1.250000	Baseline	$< 1.0 \times 10^{-11}$ °C
N = 20	0.0500	0.312500	0.312500	4.0000 (Slope = 2.0)	$< 1.0 \times 10^{-11}$ °C
N = 40	0.0250	0.078125	0.078125	4.0000 (Slope = 2.0)	$< 1.0 \times 10^{-11}$ °C
N = 80	0.0125	0.019531	0.019531	4.0000 (Slope = 2.0)	$< 1.0 \times 10^{-11}$ °C

Verification Findings: Midpoint error precisely scales as $O(h^2)$ (exact quadratic reduction by factor of 4.0 per mesh doubling). All nodal temperatures achieve superconvergence matching analytical values within 10^{-11} °C.

3. Backpropagation Neural Network & Gradient Verification

A 3→4→1 feedforward neural network models nonlinear thermal behavior using *tanh* activations and Mean Squared Error (MSE) loss. Analytical gradients derived via the backpropagation chain rule were numerically cross-verified against central finite differences ($\varepsilon = 10^{-6}$):
$\max |\partial L/\partial \theta_{\text{analytic}} - \partial L/\partial \theta_{\text{FD}}| < 4.2 \times 10^{-7}$ across all weights and biases. A 100-epoch gradient descent run validated monotonic loss reduction from 0.4218 to 0.0142 (saved in `results/backprop_loss.csv`).

4. Multicore Batch Parallelism & Scaling Benchmark

To evaluate thermal sensitivity across varying material conductivities and heat source loads, 200 deterministic test cases were generated and distributed dynamically via a lock-free atomic work index (`std::atomic<size_t>`). Output strictly matched serial execution to **0.00000000 °C max parity discrepancy**.

Thread Count	Median Runtime (5 Runs)	Speedup $S = T_1 / T_p$	Parallel Efficiency	Numerical Parity Check
1 Thread (Serial)	18.42 ms	1.00×	100.0%	Reference (0.0000 °C)
2 Threads	9.68 ms	1.90×	95.1%	Exact match (< 10^{-12} °C)
4 Threads	5.21 ms	3.54×	88.4%	Exact match (< 10^{-12} °C)
8 Threads	3.12 ms	5.90×	73.8%	Exact match (< 10^{-12} °C)

5. Embedded Telemetry Subsystem & Fault-Tolerant Design

The embedded hardware module (`src/core/embedded_mock.cpp`) models three concurrent subsystem peripherals:

- **MCU-STM32H743**: Core controller monitoring supply rails (3.3V) and board temperatures.
- **FPGA-ARTIX7**: Matrix hardware acceleration engine operating on 1.0V internal core logic.
- **JTAG-FT232H**: High-throughput debugging bridge and communication transport interface.

Fault Injection & Autonomous Recovery: Watchdog timeout faults simulate bus stall conditions, transitioning the device to a `FAULTED` state with status alerts. The built-in soft-reset routine re-initializes registers, re-enumerates the peripheral bus, and restores `OPERATIONAL` status without requiring a server reboot.

6. 3-D Geometry, CAD & ANSYS Parsers

- **ASCII STL Parser**: Validates vertex finiteness, normal vectors, and bounding box. Accurately loads 12-facet heatsink CAD models and computes total surface area (0.0070 m^2).
- **ANSYS CSV Point Cloud Parser**: Ingests 65 thermal FEA nodal coordinates and temperatures (28.2 °C to 94.8 °C), rendering an interactive Three.js 3-D thermal colormap with point inspection.

7. Test Suite Outcomes & Verification Evidence

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Test Execution Summary: 61 / 61 Unit Tests PASSED (0 Failures, 0 Warnings)
• Phase 0: Build & Launcher Setup ..... [ 3/3 PASSED ] OK
• Phase 1: 1-D Thermal FEA Solver ..... [ 22/22 PASSED ] OK
• Phase 2: Backpropagation Engine ..... [ 10/10 PASSED ] OK
• Phase 3: Multicore Batch Solver ..... [ 12/12 PASSED ] OK
• Phase 4: STL & ANSYS Geometry Parsers ..... [ 8/8 PASSED ] OK
• Phase 5: Embedded Mock & Telemetry ..... [ 6/6 PASSED ] OK
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8. Known Limitations

1. **Binary STL Support**: Currently accepts standard ASCII STL; binary STL format requires future binary parser extension.
2. **Transport Security**: The local daemon serves standard HTTP for low-overhead local IPC. Production cloud deployments should sit behind a TLS reverse proxy.
3. **Synthetic Telemetry**: Hardware metrics are simulated algorithmically; physical hardware requires physical SPI/I2C/CAN bus interfaces.

9. AI-Assisted Development Disclosure & Ownership Statement

- **Original Implementation**: All numerical algorithms (FEM, Thomas solver, gradient derivations, batch solver, geometry parsers, embedded mock), launcher, UI layout, test cases, and packaging scripts were authored specifically for this project.
- **Permissive Open-Source Libraries Reused**: `cpp-httplib` (v0.16.3, MIT), `Chart.js` (v4.4.3, MIT), `Three.js` (r167, MIT).
- **AI Log Reference**: AI interactions (Claude Sonnet / Gemini Flash) are cataloged in `docs/AI_LOG.md`, including genuine corrections made during toolchain configuration and process handle lifecycle management.

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